

COMMON DISEASES

1.1 Introduction to the unit of learning

This unit specifies the competencies required to demonstrate the knowledge of common diseases. It involves identifying the stages of disease development, demonstrating the knowledge of communicable diseases, non-communicable diseases and management of common diseases

1.2 Summary of Learning Outcomes

1. Identify the stages of disease development
2. Demonstrate the knowledge of communicable diseases
3. Demonstrate the knowledge of non-communicable diseases
4. Demonstrate the knowledge on management of common diseases

1.2.0 Learning outcome

2 1.2.1 Identify the stages of disease development

1.2.1.1 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to identify the stages of disease development. It includes Meaning of terms, Types of diseases, Natural history of a disease and Stages of disease development

1.2.1.2 Performance Standard

1. 1Common diseases are identified as per WHO epidemiological patterns
1. 2Common diseases are classified as per WHO classification guidelines
- 1.3 Natural history of common diseases is identified as per WHO guidelines

1.2.1.3 Information Sheet

1.2.1 .3.1 Definitions of terms

- i. Endemic:** The persistent presence of a disease within a geographical area in low to moderate levels not enough to deplete the pool of susceptible persons.
- ii. Hyperendemic:** Persistently high level of occurrence of a disease.
- iii. Sporadic:** An irregular pattern of occurrence of with occasional cases occurring at irregular intervals.
- iv. Epidemic/ outbreak:** The occurrence of a disease within an area which is clearly in excess of the expected level for a given time period.
- v. public health practioners prefer the term outbreak because it is less alarming/provocative.

- vi. **Pandemic:** An epidemic that is spread over several countries or continents and affects large numbers of people.
- vii. **Acute:** Develops rapidly and runs its course quickly e.g. measles and influenza
- viii. **Chronic:** Develops slowly and less severe because a weakened but viable organism is responsible. It persists for long indeterminate period of time e.g. TB and Leprosy.
- ix. **Latent:** Characterized by periods that is free from symptoms and signs. No active multiplication of agent is taking place e.g. syphilis.
- x. **Localized:** Confined to specific area e.g. boils.
- xi. **Systemic/Generalized:** Affecting the entire body with pathogens widely distributed in many tissues e.g. septicaemia (growth of bacteria throughout the circulatory system).
- xii. **Inapparent/subclinical:** Not producing symptoms in the host at a particular time.
- xiii. **Symptoms:** Disease characteristics observed or felt by the patient only
- xiv. **Sign:** Disease characteristic observed by examining through visual appreciation or using an instrument.
- xv. **Syndrome:** A combination of signs and symptoms occurring together in an illness. Syndrome can be used in absence of clinical test to diagnose disease and institute appropriate action.
- xvi. **Sequelae:** After effects left by certain diseases e.g. paralysis after infection with polio virus.

1.2.1.3.2 Types of diseases

Communicable, or infectious diseases, are caused by microorganisms such as bacteria, viruses, parasites and fungi that can be spread, directly or indirectly, from one person to another. Some are transmitted through bites from insects while others are caused by ingesting contaminated food or water.

A variety of disease-producing bacteria and viruses are carried in the mouth, nose, throat and respiratory tract. Conditions such as leprosy, tuberculosis (TB) and different strains of influenza (flu) can be spread by coughing, sneezing, and saliva or mucus on unwashed hands.

Sexually transmitted infections (STIs) such as HIV and viral hepatitis are spread through the exposure to infective bodily fluids such as blood, vaginal secretions and semen. Hepatitis is a significant concern in the African Region and the majority of people living with hepatitis B and C are unaware of their infections.

Insects play a significant role in the transmission of disease. Bites from *Anopheles* mosquitoes transmits malaria parasites that can wreak havoc on high-risk populations such as children under age 5 and pregnant women. Yellow fever has also seen

resurgence due to reduced vaccination efforts. Many neglected tropical diseases are caused by unsafe water, poor housing conditions and poor sanitation in the Region.

1.2.1.3.3 List of Communicable Diseases

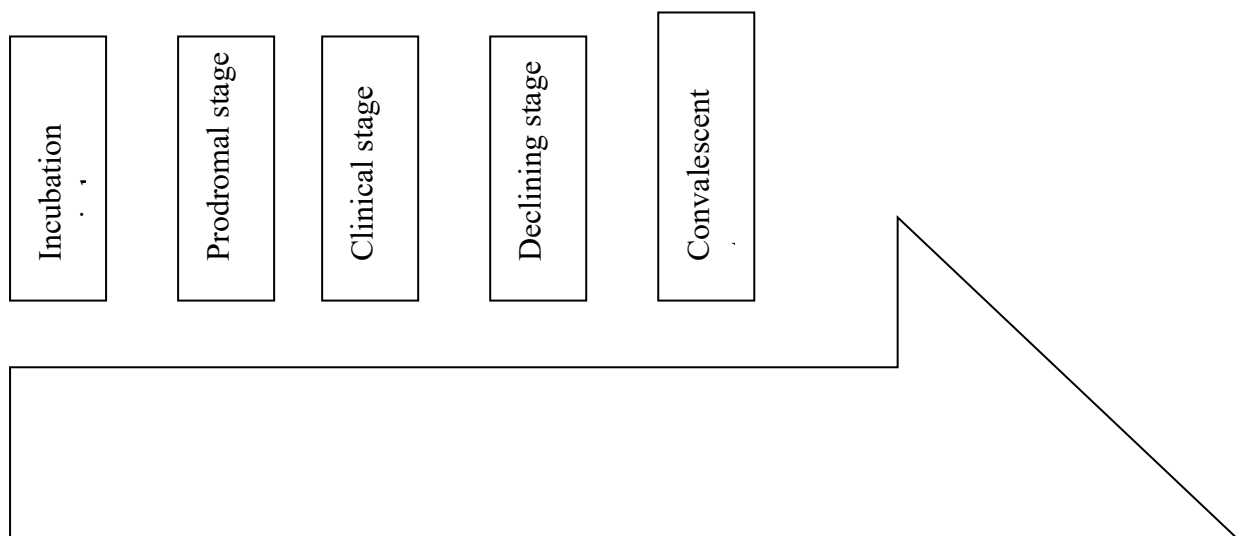
The diseases below are among them.

- ✓ 2019-nCoV
- ✓ Ebola
- ✓ Flu
- ✓ Hepatitis A
- ✓ Hepatitis B
- ✓ HIV/AIDS
- ✓ Measles
- ✓ Pertussis
- ✓ Rabies
- ✓ Sexually Transmitted Disease
- ✓ Shigellosis
- ✓ Tuberculosis
- ✓ West Nile Virus
- ✓ Zika

1.2.1.3.4 Natural history of a disease:

- Refers to the progress of a disease process in an individual without intervention over time including complications, symptoms change, remissions, disability cure recovery and death.
- The process begins with exposure to or accumulation of factors capable of causing disease in a susceptible host.
- Most diseases have a characteristic natural history although the time frame and the specific manifestations of the disease vary from person to person.

1.2.1.3.5 Stages in the natural history of a disease:



Recovery

Period in which illness is
apparent

Latent period
nicability

Period of commu-
Carrier

1.2.1.3.6 Incubation Period:

- ❖ Pathogen has embedded in the body of a susceptible host.
- ❖ No signs and symptoms are apparent.
- ❖ Micro-organisms have invaded the host and are migrating to various tissues causing some pathological changes.
- ❖ The pathogens have not increased to sufficient numbers to produce enough toxins to produce discomfort nor cause the individual to be infective.
- ❖ The period is usually variable in length and usually disease specific.

1.2.1.3.7 Prodromal stage:

- ❖ First symptoms of the disease appear.
- ❖ Patient is aware of discomfort but no precise signs/symptoms to permit diagnosis
- ❖ Patient highly contagious to others.
- ❖ Host immune responses become operative.
- ❖ Marks transition from sub-clinical to clinical disease.

1.2.1.3.8 Clinical stage:

- ❖ Period of illness
- ❖ Characteristic disease symptoms occur.
- ❖ Most diagnoses are made during this stage.
- ❖ During the acute stage of this period, the patient is sufficiently ill to alter work/ school activities.
- ❖ In some people the disease process may never progress to clinically apparent illness.
- ❖ In others the disease process may result in a wide spectrum of clinical illness, ranging from mild to severe or fatal.
- ❖ Chemotherapeutic intervention or adequate immune response may lead to recovery.

1.2.1.3.9 Decline stage:

- ❖ Period of decline if disease is not fatal

- ❖ First sign of recovery manifests i.e. disappearance of signs and symptoms.
- ❖ Disease starts to end.
- ❖ Disease becomes latent or intermittently recurs.

1.2.1.3.10 Convalescent stage:

- ❖ Disease ends
- ❖ Progresses to carrier stage or freedom from pathogen.
- ❖ Body's defense may provide lifelong immunity.

1.2.1.3.11 Full recovery:

- ❖ Marks the end of disease syndrome and return to full health.

3 1.2.2 Demonstrate the knowledge of communicable diseases

1.2.2.1 INTRODUCTION COMMUNICABLE DISEASES

A disease that is passed from one person to another person is called a communicable or transmissible disease. Transmissible diseases include: measles, HIV infection, tuberculosis, chickenpox, gonorrhoea, scabies, malaria, cholera, and roundworms among others. Communicable diseases are among the most important diseases in this country. They are important because:

- ✓ Many of them are common
- ✓ Some of them are very serious and cause death and disability
- ✓ Some of them cause widespread outbreaks of disease – epidemics
- ✓ Most of them can be prevented by fairly simple means

In this section you will cover disease patterns in the community, the meaning of host and infection, as well as the transmission cycle of communicable diseases.

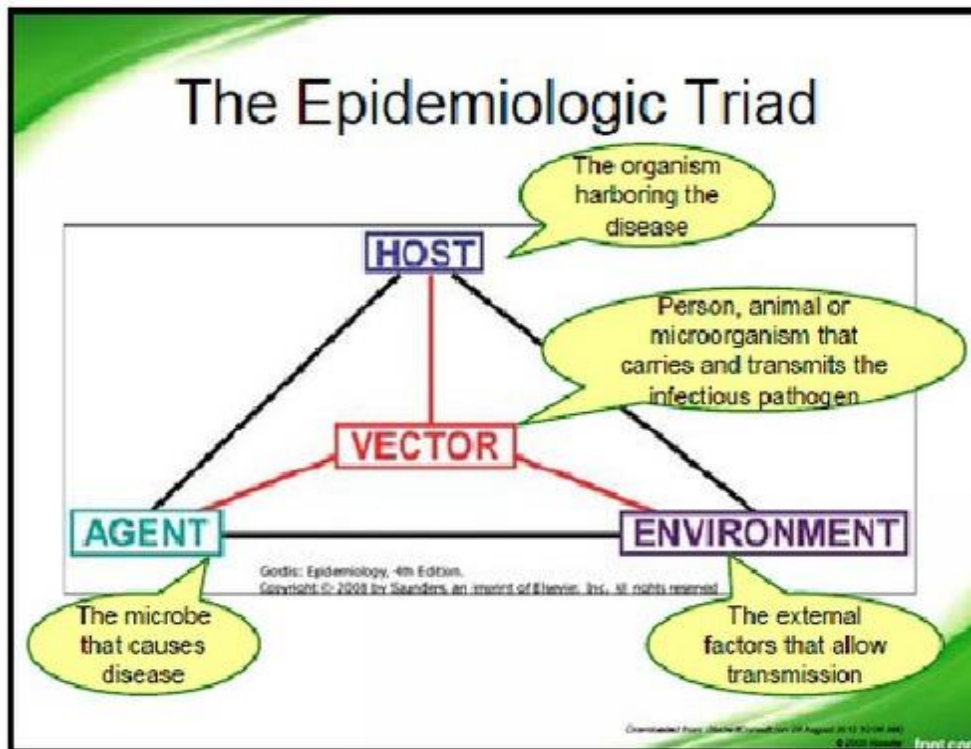
1.2.2.1.1 Patterns of Communicable Diseases in the Community

Different diseases are common in different places and at different times. To understand why this happens, you need to consider the disease causative organisms (the agents); the people they infect (the hosts); and the surroundings in which they live (the environment).

A delicate balance exists between the agent, the host and the environment and it can change in different ways. For instance, the agent needs a suitable environment in which to grow and multiply and thus be able to spread and infect other hosts. If the environment does not support the agent it dies or transforms to a dormant state.

The host (person) is also affected by the environment. For example, a person may live in a hot, wet climate where there are many mosquitoes. However they can change this environment by draining swamps, clearing the vegetation and adding competing hosts such as animals. If the balance is shifted against the agent, the disease will be controlled and the number of cases will go down.

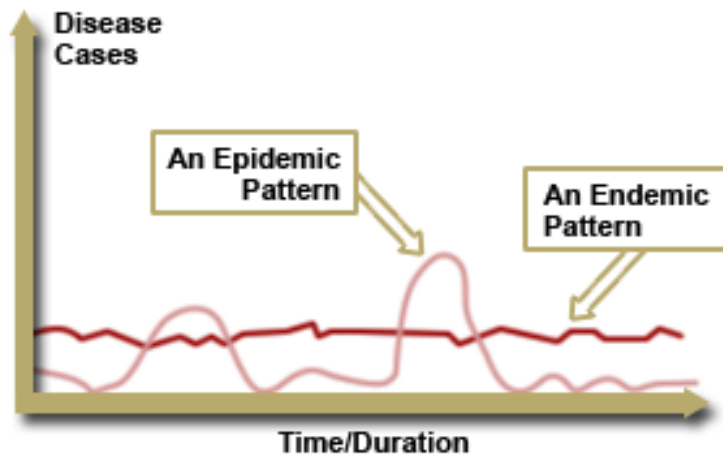
1.2.2.1.2 EPIDEMIOLOGIC TRIANGLE



When the balance between the agent, the host and the environment is fairly constant, you tend to see approximately the same number of cases of the disease every month. When this happens the disease is said to be endemic. When the balance is shifted in favour of the agent (organism), for example, when many non immune children have been born in an area since the last measles epidemic, a large number of cases of measles may occur in a short time. This is called an epidemic. Epidemic diseases occur during certain periods or seasons and cause sudden deaths and much suffering in the community.

An endemic disease can be termed as that which occurs in a given population at a constant rate over a period of several years. An epidemic disease is that which occurs in a population at a higher rate than is usually the normal for that population over a given time interval.

Diagrammatically the endemic and epidemic disease patterns can be illustrated as follows:-



1.2.2.1.3 Some Common Epidemic Diseases in Kenya

- ✓ Cholera
- ✓ Typhoid fever
- ✓ Highland malaria
- ✓ Acute bacterial meningitis

In Kenya, malaria is endemic in the lowlands, such as the Tana River basin, the coastal strip, and the Lake Victoria region. Schistosomiasis which is related to water use is endemic around the Lake Victoria region and the Mwea irrigation scheme. Leishmaniasis is endemic in Baringo, along Tana River, and along the River Athi in Machakos.

In some parts of the country, some disease outbreaks occur only occasionally without a regular pattern. Such diseases are said to be sporadic in their occurrence.

1.2.2.1.4 The Host and Infection

At this stage, infection does not produce clear signs and symptoms. The host's immune system is trying to fight off the agent. In some cases, the organism is overcome by the host immune cells hence no signs and symptoms are felt and the infection process is terminated.

1.2.2.1.5 Clinical Infection

This is the period when the host develops detectable symptoms and signs of an illness. At this time the agent has multiplied within the host overcoming the host's immune system and has started causing abnormal functioning of some body cells and tissues. This produces overt signs and symptoms of the disease.

It is important for you to understand these stages because people with symptoms are easier to identify as they come to your health facilities for treatment. People with sub-clinical infections do not always know they are infected and hence are a danger to other people. They are also difficult to detect in the general population without special tests. An individual who

is suffering from a sub-clinical infection is also likely to infect others, as in the case of HIV infection which leads to AIDS after a long period.

They are therefore known as carriers. An individual who develops a clinical or sub-clinical infection is said to be susceptible to the disease. A susceptible individual is one whose body lacks resistance to the disease. Resistance of the body to a disease occurs due to various immunity mechanisms.

1.2.2.1.6 Modes of transmission of communicable diseases

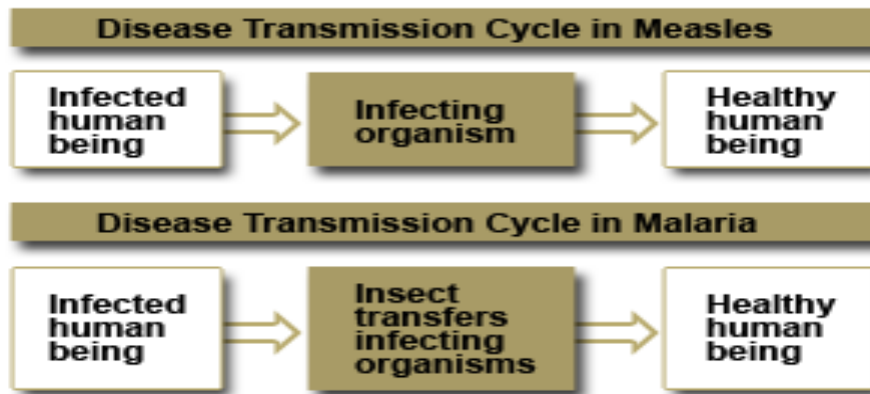
- ✓ Direct contact, for example sexual contact, contact with skin or mucous membranes
- ✓ Vectors
- ✓ Faecal-oral (ingesting contaminated food and water)
- ✓ Airborne
- ✓ Tran placental (mother to foetus)
- ✓ Blood contact (transfusion, surgery, injection)
- ✓ Contact with animals or their products

1.2.2.1.7 The Disease Transmission Cycle

A disease transmission cycle is a series of steps that a disease-causing organism undergoes in its disease-causing process. Disease-causing organisms are living things that need somewhere to live and reproduce. This may be within inanimate or animate environment (example in rodents, insect, or the human body), which is known as the reservoir of infection. The human being is the main reservoir of most of the communicable diseases that affect humanity.

When an infection spreads to a new host, the place, animal or human from which the organism came from is called the source of the infection. The way in which an organism leaves the source (the infected host) and passes to a new susceptible host is called the route of transmission. Each disease-causing organism has particular routes which play a large part in how these organisms spread in the community.

For example, some organisms are spread through water and food, while others are spread by vectors like mosquitoes and snails. The diagram below illustrates the differences between the disease transmission cycle in measles, and in malaria where the causative organism passes from human being to mosquito and back to the human being.



Every transmission cycle is made up of three parts:-

1.2.2.1.7.1 The Source

This is where the disease-causing organisms spread from. It could be an infected person, animal, place, or object. The reservoir is the source of infection.

1.2.2.1.7.2 Transmission Route

The main routes of transmission are:

- Direct contact, for example sexual contact, contact with skin or mucous membranes
- Vectors
- Faecal-oral (ingesting contaminated food and water)
- Airborne
- Tran placental (mother to foetus)
- Blood contact (transfusion, surgery, injection)
- Contact with animals or their products

1.2.2.1.7.3 Susceptible Host

An individual who has low resistance to a particular disease is said to be a susceptible host for that disease. There are a number of factors which lower the body's resistance to a disease:

- ✓ Not having come in contact with the disease-causing organisms before and therefore not having any immunity to it. For example, passive immunity against measles is lost at the age of 6 - 12 months. Therefore if a child comes into contact with the measles virus after this age, they may develop the disease.
- ✓ Having a serious illness like AIDS which suppresses a person's immunity. People with AIDS have a high risk of developing tuberculosis.
- ✓ Malnutrition
- ✓ Certain drugs such as those used to treat cancer can lower a person's resistance to disease.

4 1.2.3 Demonstrate the knowledge of non-communicable diseases

1.2.3.1 Meaning of terms

1.2.3.1.1 What is a Non-Communicable Disease?

- ▶ A non-communicable disease are diseases that cannot be spread from person to person.

- ▶ Some non-communicable diseases are chronic.

Chronic means that it is present either continuously or on and off over a long period.

- ▶ Multiple Sclerosis is an example of a degenerative disease.

Degenerative disease is a disease that causes breakdown in body cells, tissues, and organs as it progresses

1.2.3.1.2 Diseases Present at Birth

- ▶ Some congenital diseases are caused by heredity, while others are caused by a mother's choice of lifestyle while pregnant.

- ▶ Congenital Disorders are all disorders that are present when the baby is born

- ▶ Heredity is the passing of traits from parents to their children.

- ▶ Examples of traits that can be passed from parents to their children are: eye color, sickle cell anemia, etc.

1.2.3.2 Risk factors of non-communicable diseases

1.2.3.2.1 Lifestyle choices and Disease

× Factors that You Cannot Control:

- a) Heredity
- b) Age
- c) Gender
- d) Ethnic Group

1.2.3.2.2 Lifestyle choices and Disease

× Factors that you Can Control:

- i. Eat healthy food
- ii. Stay physically active
- iii. Maintain a healthy weight
- iv. Get enough sleep
- v. Manage stress
- vi. Avoid tobacco, alcohol and other drugs

1.2.3.2.3 Environmental Factors and Disease

- ▶ Many substances in the environment can cause serious health problems (ex. lead in fish)
- ▶ Carbon Monoxide in high levels can cause illness and death
- ▶ Smog can cause respiratory / breathing problems in some people

1.2.3.2.4 Common Non-Communicable Diseases

Some non-communicable diseases are more common than others. The four main types of non-communicable diseases include cardiovascular disease, cancer, chronic respiratory disease, and diabetes.

1.2.3.2.4 .1 Cardiovascular disease

Poor diet and physical inactivity can cause increased:

- ✓ blood pressure
- ✓ blood glucose
- ✓ blood lipids
- ✓ obesity

These conditions increase the risk of developing cardiovascular disease. Some people are born with (genetically predisposed to have) certain cardiovascular conditions.

Cardiovascular disease is the leading cause of noncommunicable disease deaths. Some common noncommunicable cardiovascular conditions and diseases include:

- ✓ heart attack
- ✓ stroke
- ✓ coronary artery disease
- ✓ cerebrovascular disease
- ✓ peripheral artery disease (PAD)
- ✓ congenital heart disease
- ✓ deep vein thrombosis and pulmonary embolism

1.2.3.2.4 .2 Cancer

Cancer affects people of all ages, socioeconomic statuses, genders, and ethnicities. It's the second most-common cause of noncommunicable disease death globally.

Some cancers cannot be avoided due to genetic risks. However, the World Health Organization estimates that of cancers are preventable with adoption of healthy lifestyle choices.

Key steps in preventing disease include:

- ✓ avoiding tobacco
- ✓ limiting alcohol
- ✓ getting immunized against cancer-causing infections

The most common cancer deaths in men worldwide include:

- ✓ lung
- ✓ liver
- ✓ stomach
- ✓ colorectal
- ✓ prostate

The most common cancer deaths in women worldwide include:

- ✓ breast
- ✓ lung
- ✓ colorectal
- ✓ cervical
- ✓ stomach

1.2.3.2.4 .3 Chronic respiratory disease

Chronic respiratory diseases are ailments affecting the airways and lung structures. Some of these diseases have a genetic basis. However, other causes include lifestyle choices such as smoking and environmental conditions like exposure to air pollution, poor air quality, and poor ventilation. While these diseases are incurable, they can be managed with medical treatment. The most common chronic respiratory diseases include:

- ✓ chronic obstructive pulmonary disease (COPD)
- ✓ asthma
- ✓ occupational lung diseases, such as black lung
- ✓ pulmonary hypertension
- ✓ cystic fibrosis

1.2.3.2.4 .4 Diabetes

Diabetes occurs when the body cannot produce enough insulin, a hormone that regulates blood sugar (glucose). It can also occur when the body cannot effectively use the insulin it produces. Some effects of diabetes include heart disease, vision loss, and kidney injury. If blood sugar levels are not controlled, diabetes can seriously damage other organs and systems in the body over time. There are two main types of diabetes:

- **Type 1 diabetes** is often diagnosed during childhood or young adulthood. It's the result of an immune system dysfunction.
- **Type 2 diabetes** is often acquired during later adulthood. It's typically the result of poor diet, inactivity, obesity, and other lifestyle and environmental factors.

Other types of diabetes include:

- **gestational diabetes**, which causes elevated blood sugar in 3 to 8 percent of pregnant women in the United States
- **prediabetes**, a condition defined by higher-than-normal blood sugar levels that lead to a very high risk of developing type 2 diabetes in the near future

1.2.4.1 PRINCIPLES (METHODS) OF COMMUNICABLE DISEASE CONTROL

1.2.4.1.1 Methods of Communicable Disease Control

Communicable disease can be controlled and eradicated from the community. When thinking about the control of diseases it is always good to think of all the possible methods. In most cases one or two specific methods will have the greatest effect and should be the focus of your activity, in other cases some methods will be useless against the disease. The aim of control is to tip the balance against the agent.

The control and eradication of communicable diseases can be done by:

- ✓ Attacking the source of the disease causing organism
- ✓ Interrupting the transmission route
- ✓ Protecting the susceptible host

You will now look at each method in turn.

1.2.4.1 .1.1 Attacking the Source

There are various specific measures which can be used to control the spread of an infectious disease.

They include:

- ✓ Treating the infected person or animal with the appropriate antibiotics that destroy the disease causing-organism.
- ✓ Treating the carriers and sub-clinical cases after carrying out screening tests among suspected individuals or groups.
- ✓ Treating specific groups of persons who are at high risk of being infected (mass treatment). This is called chemoprophylaxis.
- ✓ Isolating those persons who are infected with highly infectious diseases such as ebola, marburg fever, lassa fever; so as to prevent the spread of the organism to other healthy people.
- ✓ Treating sick animals such as cattle suffering from brucellosis, immunizing animals such as cows from anthrax, and dogs from rabies; killing sick animals such as rats to control plague and dogs to prevent rabies; separating humans from animals.
- ✓ Notifying the local health authorities immediately you suspect a patient is suffering from an infectious disease.

Though this does not directly affect the source, it is an essential way of keeping watch on the number of new cases and thereby monitoring the effectiveness of the control programme. All of the methods mentioned on the previous page are methods of controlling the reservoir - where an animal is the reservoir.

In summary you can state that the measures for attacking the source are:

- ✓ Treating the infected person/s
- ✓ Treating the carrier
- ✓ Mass treatment of persons at risk
- ✓ Isolating the infected person/s
- ✓ Treating the sick animal such as cows
- ✓ Immunizing animals such as dogs and cattle
- ✓ Killing the animal reservoir such as rats
- ✓ Separating humans and animals

1.2.4.1 .1.1 Interrupting the Transmission Cycle

A number of methods are used to interrupt the transmission cycle. They include the following:

- ✓ Personal hygiene
- ✓ Environmental health
- ✓ Water and sanitation
- ✓ Vector control
- ✓ Good and adequate housing
- ✓ Effective food handling and adequate nutrition

NB: Sterilization of medical equipment and the use of sterile surgical equipment (These methods are useful for interrupting the transmission of diseases such as, Human Immunodeficiency Virus (HIV) infection and hepatitis-B infection.

Remember: A clean environment and good personal hygiene are the most important measures in the primary prevention of diseases.

1.2.4.1 .1.1 Protecting the Host

This is the third principle of controlling the spread of communicable disease in the community. Any person who is not yet infected by a specific disease-causing organism is known as a susceptible host. This is because they are at risk of contracting the infection. All susceptible hosts must be protected from contracting the infection.

Remember: The most effective way of controlling communicable diseases is to use a combination of methods: attacking the source of the infecting organism, interrupting the route of transmission, and protecting the susceptible host.

1.2.4.2 Basic management of common diseases

There are various specific and general measures for protecting the host.

1.2.4.2.1 Specific Measures

- ✓ Immunization using vaccines such as the KEPI vaccine
- ✓ Chemoprophylaxis using for example:
- ✓ Proguanil (PaludrineR) to suppress malaria parasites
- ✓ Tetracycline during cholera outbreaks
- ✓ Cotrimoxazole during plague Outbreaks

1.2.4.2.2 General Measures

- ✓ Use of barriers such as bed nets, gowns, gloves to prevent insect bites (especially mosquitoes)

- ✓ Use of chemicals for example insect repellents to prevent mosquito bites
- ✓ Wearing shoes to prevent penetration by hookworms from the soil
- ✓ Adequate housing to reduce overcrowding
- ✓ Improved nutrition
- ✓ Adequate ventilation
- ✓ Health education

1.2.4.2.3 Other Control Measures

There are other useful measures that can be taken to control the spread of communicable disease. Among these is the notification of disease. Notification requires you to keep watch (surveillance) on the number of new cases of communicable diseases in your area of work and to immediately inform the local health authority when you come across a patient suffering from an infectious disease. One of the main reasons for notification is to help the health authorities take measures to confirm your suspicion and to control the spread of the disease. Notification of infectious communicable diseases is the responsibility of all health care workers. It is also a legal requirement according to the Public Health Act, Chapter (cap) 242; section eight of the laws of Kenya.

Remember: It is your responsibility to notify your local health authority immediately should you suspect the presence of an infectious disease.

1.2.4.3 List any six notifiable diseases found in Kenya.

1.2.4.3.1 Notifiable Diseases in Kenya

- Plague
- Cholera
- Measles
- Poliomyelitis
- Diphtheria
- Tuberculosis
- Anthrax
- Trypanosomiasis
- Typhoid fever
- Whooping cough
- Meningococcal meningitis
- Rabies

Yellow fever

The diseases in bold spread so quickly that they need international control measures. These diseases are reported by the Ministry of Health to the World Health Organization (WHO).

Application of Communicable Disease Control Measures

The actual application of the control methods you have just seen can be undertaken by different groups of people and institutions at various levels. These include individuals and village level, dispensary and health centre level and the district and central government (Ministry of Health) level.

Remember: A successful communicable disease control program is the one that involves members of the community.

Further reading

Communicable diseases 4th Edition by Erick Nordberg

Basic epidemiology by r.Beaglehole

1.2.4.4 Learning Activities

Group discussions

1.2.4.4.1 Practical activities

Task: *S ward in Y Sub County has been the most affected area with Communicable and non-communicable disease cases and high mortality rate. You have been assigned as the community health worker of that particular area. Carry out a survey on how you will curb that problem and improve the health standards of that community.*

1.2.4.4.2 A case study

Typhoid fever is a disease that manifests clinically with high fever and headache. Suppose Abebe is infected with the infectious agent of typhoid fever, but he has no manifestations of the disease. He works in a cafe and among 20 people he served in one day, five got infected, but only three of these developed the disease. Among the three who developed typhoid fever, two recovered and one died.

1. What are the likely modes of transmission?
2. Which of the affected persons are active cases and which are carriers?
3. Can you group the 20 people who were served in the cafe into the four stages of the natural history of a communicable disease?